tsvc/s211 · s211.c · bare_llm · claude-haiku-4-5-20251001 · BROKEN

−0 / +19 lines of 176 · 2 parallel region(s) added · 0 of 0 candidate(s) kept · RESTRUCTURING (2 code line(s) added, 0 removed) · restructuring level 0

BEFORE

```
132  }
133
134  static real_t kernel_s211(void)
135  {
136      for (int nl = 0; nl < R; nl++) {
137          for (int i = 1; i < LEN_1D-1; i++) {
138              a[i] = b[i - 1] + c[i] * d[i];
139              b[i] = b[i + 1] - e[i] * d[i];
140          }
141          pb_mix(nl);
142      }
143      return (real_t)0;
144  }
145
```

AFTER

```
132  }
133
134  static real_t kernel_s211(void)
135  {
136      for (int nl = 0; nl < R; nl++) {
137          /* Phase 1: Compute all a[i] values in parallel.
138           * Each iteration i reads b[i-1] (which is never written in this loop)
139           * and writes a[i]. No loop-carried dependencies.
140           * i: loop variable (shared/implicit)
141           * a, b, c, d: shared arrays (read-only for b,c,d; write-only for a)
142           */
143          #pragma omp parallel for
144          for (int i = 1; i < LEN_1D-1; i++) {
145              a[i] = b[i - 1] + c[i] * d[i];
146          }
147
148          /* Phase 2: Compute all b[i] values in parallel.
149           * Each iteration i reads b[i+1] (to the right) and writes b[i] (to th
150           * Iteration j>i reads from j+1 (> i), so it doesn't depend on iterati
151           * write to i. All reads occur at higher indices than all writes. Para
152           * i: loop variable (shared/implicit)
153           * b, d, e: shared arrays (read-only for d,e; read/write for b)
154           */
155          #pragma omp parallel for
156          for (int i = 1; i < LEN_1D-1; i++) {
157              b[i] = b[i + 1] - e[i] * d[i];
158          }
159
160          pb_mix(nl);
161      }
162      return (real_t)0;
163  }
164
```

file

red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is